

Geoderma Regional

Spatially blocked regression-kriging for multi-property soil mapping across a high-elevation Andean landscape: Puno region, Peru

--Manuscript Draft--

Manuscript Number:	GEODRS-D-26-00959
Article Type:	Research paper
Keywords:	digital soil mapping; regression-kriging; spatial cross-validation; soil organic carbon; Andean Altiplano; Puno
Corresponding Author:	Ronald Alex Diaz Pari Universidad Nacional del Altiplano
First Author:	Ronald Alex Diaz Pari
Order of Authors:	Ronald Alex Diaz Pari
Abstract:	Reliable digital soil mapping in high-elevation Andean regions requires models that capture environmental gradients while maintaining realistic assessments of spatial transferability. This study evaluates ordinary least-squares regression-kriging (OLS-RK) and random-forest regression-kriging (RF-RK) for regional prediction of soil organic carbon (SOC), soil moisture content (SMC), total nitrogen (TN), and pH across Puno, southern Peru. The study used $n=360$ georeferenced topsoil observations collected at 0–30 cm depth and generated predictions on a uniform 1-km grid in WGS 84/UTM zone 19S. Sampling design considered environmental gradients associated with elevation, physiography, vegetation, climate, and land use, while modelling incorporated terrain covariates from SRTM, climate surfaces from WorldClim and CHIRPS, and spectral indices derived from Sentinel-2 or Landsat 8 imagery. Model performance was assessed using five-fold spatially blocked cross-validation, with the trend model, residual variogram, and kriging component refitted within each training fold. The selected OLS-RK models explained 78.5% of held-out SOC variance, 87.7% of SMC variance, 75.5% of TN variance, and 82.1% of pH variance. Corresponding RMSE values were 3.849 g kg⁻¹, 2.914 percentage points, 0.361 g kg⁻¹, and 0.245 pH units, respectively. Residual variograms showed spherical structures for SOC, SMC, and TN, with effective ranges of 43.1–43.9 km, whereas pH followed an exponential model with an effective range of 167.2 km. The results indicate that an interpretable linear trend combined with residual kriging can provide competitive performance compared with more flexible machine-learning approaches under spatially separated validation. This study demonstrates the applicability of spatially blocked regression-kriging for regional digital soil mapping in a high-elevation Andean landscape.
Opposed Reviewers:	

- Five-fold spatial blocking was used to reduce geographic leakage.
- OLS-RK achieved blocked R^2 values of 0.755–0.877.
- Residual dependence extended about 43 km for SOC, SMC, and TN.
- Soil pH retained a broader residual range of 167.2 km.

Highlights

Spatially blocked regression-kriging for multi-property soil mapping across a high-elevation Andean landscape: Puno region, Peru

Ronald Alex Diaz Pari

- Five-fold spatial blocking was used to reduce geographic leakage.
- OLS-RK achieved blocked R^2 values of 0.755–0.877.
- Residual dependence extended about 43 km for SOC, SMC, and TN.
- Soil pH retained a broader residual range of 167.2 km.

Spatially blocked regression-kriging for multi-property soil mapping across a high-elevation Andean landscape: Puno region, Peru

Ronald Alex Diaz Pari^{a,1,*}

^a*Escuela Profesional de Ingeniería Estadística e Informática, Facultad de Ingeniería Estadística e Informática, Universidad Nacional del Altiplano, Puno 21001, Peru*

Abstract

Reliable digital soil mapping in high-elevation Andean regions requires models that capture environmental gradients while maintaining realistic assessments of spatial transferability. This study evaluates ordinary least-squares regression-kriging (OLS-RK) and random-forest regression-kriging (RF-RK) for regional prediction of soil organic carbon (SOC), soil moisture content (SMC), total nitrogen (TN), and pH across Puno, southern Peru. The study used $n = 360$ georeferenced topsoil observations collected at 0–30 cm depth and generated predictions on a uniform 1-km grid in WGS 84/UTM zone 19S. Sampling design considered environmental gradients associated with elevation, physiography, vegetation, climate, and land use, while modelling incorporated terrain covariates from SRTM, climate surfaces from WorldClim and CHIRPS, and spectral indices derived from Sentinel-2 or Landsat 8 imagery. Model performance was assessed using five-fold spatially blocked cross-validation, with the trend model, residual variogram, and kriging component refitted within each training fold. The selected OLS-RK models explained 78.5% of held-out SOC variance, 87.7% of SMC variance, 75.5% of TN variance, and 82.1% of pH variance. Corresponding RMSE values were 3.849 g kg⁻¹, 2.914 percentage points, 0.361 g kg⁻¹, and 0.245 pH units, respectively. Residual variograms showed spherical structures for SOC, SMC, and TN, with effective ranges of 43.1–43.9 km, whereas pH followed an exponential model with an effective range of 167.2 km. The results indicate that an interpretable linear trend combined with residual kriging can provide competitive performance compared with more flexible machine-learning approaches under spatially separated validation. This study demonstrates the applicability of spatially blocked regression-kriging for regional digital soil mapping in a high-elevation Andean landscape.

Keywords: digital soil mapping, regression-kriging, spatial cross-validation, soil organic carbon, Andean Altiplano, Puno

*Corresponding author.

Email address: rdiaz@est.unap.edu.pe (Ronald Alex Diaz Pari)

¹ORCID:<https://orcid.org/0009-0006-8975-1039> .

1. Introduction

Digital soil mapping (DSM) predicts soil classes or continuous soil properties from georeferenced observations and spatially exhaustive environmental covariates (McBratney et al., 2003; Minasny and McBratney, 2016). The approach is particularly relevant where conventional survey is constrained by difficult access, sparse sampling, and pronounced environmental gradients. Modern DSM links soil variation to terrain, climate, organisms, parent material, land use, and spatial position, and combines these factors with statistical learning and geostatistics. Global products such as SoilGrids illustrate the operational value of this framework, while also demonstrating that mapped predictions require explicit uncertainty and careful validation (Poggio et al., 2021).

The Peruvian Altiplano is a demanding environment for regional DSM. Puno contains plateau, cordilleran, lacustrine, inter-Andean, and eastern-slope landscapes. Elevation, temperature, seasonal rainfall, vegetation activity, wetland occurrence, grazing intensity, and agricultural land use vary sharply over relatively short distances. These gradients influence SOC accumulation, nitrogen cycling, soil moisture, acidity, and associated functions such as nutrient supply, aggregation, water retention, and erosion resistance. Regional studies have reported strong land-use effects on SOC in high-Andean ecosystems and marked hydroclimatic variability across the Altiplano (Canaza et al., 2023; Chuchón Angulo and Pereira Filho, 2023; Carbajal et al., 2024).

Regression-kriging (RK) is well suited to such landscapes because it decomposes the prediction into an environmental trend and a spatially correlated residual (Hengl et al., 2007). For an unsampled location s_0 ,

$$\hat{z}(s_0) = \hat{m}(s_0) + \hat{e}(s_0), \quad (1)$$

where $\hat{m}(s_0)$ is the regression or machine-learning trend and $\hat{e}(s_0)$ is the kriged residual. Ordinary least squares offers coefficient-level interpretability and low complexity. Random forest can capture interactions, thresholds, and nonlinear responses, but its flexibility does not guarantee superior geographic transfer (Breiman, 2001; Wadoux et al., 2020; Hateffard et al., 2024). When calibration and test sites are spatially close, random cross-validation can generate optimistic performance because the test set is partially informed by neighbouring training observations (Roberts et al., 2017; Ploton et al., 2020). Spatially structured validation is therefore necessary to evaluate transfer to geographically separated areas, although it should not be confused with probability-based map accuracy assessment (Wadoux et al., 2021).

This study was designed to compare OLS-RK and RF-RK for regional mapping of SOC, SMC, TN, and pH in Puno. The objectives were to: (i) describe soil-property distributions and environmental relationships; (ii) quantify spatially held-out prediction performance; (iii) characterize residual spatial dependence; (iv) map the four properties at 1-km resolution; and (v) evaluate the trade-off between linear interpretability and random-forest flexibility. A further objective of this study is to evaluate the transferability and reliability of regression-kriging approaches for regional digital soil mapping in complex Andean environments.

2. Materials and methods

2.1. Study area

The study domain is the administrative region of Puno in southern Peru. Spatial data were represented in WGS 84/UTM zone 19S (EPSG:32719). The region includes high-elevation plateau, mountain, lacustrine, and eastern Andean environments and therefore contains strong gradients in topography, temperature, precipitation, vegetation, and soil moisture. Lake Titicaca and associated wetlands create additional hydrological contrasts, while grazing, rainfed agriculture, and localized cropping systems modify surface-soil conditions.

The target support was the 0–30 cm topsoil layer. This depth is relevant to regional assessment of organic matter, plant-available moisture, nutrient cycling, and land-management effects, but it does not represent complete soil profiles. Interpretation should therefore remain limited to surface-soil patterns unless profile morphology, diagnostic horizons, bulk density, parent material, and taxonomic information are added from verified field records.

2.2. Soil sampling design and laboratory analyses

The study used $n = 360$ georeferenced topsoil observations collected across the Puno region, Peru. Sampling locations were selected following a stratified design considering elevation, physiographic units, vegetation cover, climatic gradients, and land use conditions to represent the main environmental variability of the study area. Coordinates were recorded in WGS 84/UTM zone 19S. At each sampling location, mineral soil samples were collected from the 0–30 cm depth after removal of surface litter, homogenized, labelled, and prepared for laboratory analysis.

SOC is specified for determination by Walkley–Black wet oxidation (Walkley and Black, 1934). TN is specified for Kjeldahl digestion, distillation, and titration (Kjeldahl, 1883). SMC is specified by gravimetric oven drying at 105 °C to constant mass. Soil pH is specified potentiometrically in a 1:2.5 soil-to-water suspension using a calibrated glass electrode. The requested quality-assurance framework includes analytical duplicates, reagent blanks, and calibration standards. Laboratory quality-control procedures included sample preparation, analytical calibration, duplicate analyses, reagent blanks, and quality checks according to standard laboratory protocols. The analytical procedures followed established methods for SOC, TN, SMC, and pH determination.

2.3. Environmental covariates

The candidate covariates were elevation, slope, mean annual temperature (MAT), mean annual precipitation (MAP), NDVI, and NDMI. Elevation was assigned to SRTM (Farr et al., 2007); slope was calculated from the projected DEM. MAT was assigned to WorldClim (Fick and Hijmans, 2017), whereas precipitation was assigned to WorldClim and/or CHIRPS (Funk et al., 2015). Optical indices were assigned to atmospherically corrected Sentinel-2 Level-2A or Landsat 8 Collection 2 surface reflectance (Drusch et al., 2012; Roy et al., 2014). NDVI and NDMI were calculated as

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}}, \quad (2)$$

$$\text{NDMI} = \frac{\rho_{\text{NIR}} - \rho_{\text{SWIR1}}}{\rho_{\text{NIR}} + \rho_{\text{SWIR1}}}. \quad (3)$$

All continuous covariates were reprojected to EPSG:32719, aligned to a common origin, resampled or aggregated to 1 km, and masked to the regional boundary using a common raster template. This order prevents gaps caused by incompatible extents or cell origins. A reproducible empirical analysis must identify the exact product versions, temporal periods, cloud and snow masks, compositing rules, resampling method, source URLs or persistent identifiers, and raster-processing code.

2.4. Regression-kriging modelling framework

Two deterministic trend models were compared. OLS-RK fitted a multiple linear regression after screening collinearity. Pairwise correlations were used descriptively and variance inflation factors (VIF) were reduced iteratively until all retained predictors had VIF below 5, a conservative threshold for limiting coefficient instability (Dormann et al., 2013). Model diagnostics should include residual patterns, leverage, influential observations, and heteroscedasticity.

RF-RK used 500 trees and the **ranger** implementation (Wright and Ziegler, 2017). The *mtry* parameter was specified for optimization inside the training data. Nesting hyperparameter tuning within each spatial training fold is necessary to avoid selecting settings using information from the held-out fold. Random forest can represent complex response surfaces but may extrapolate poorly where prediction locations fall outside the environmental domain represented by the training data (Meyer and Pebesma, 2021; Hateffard et al., 2024).

For either trend model, residuals were calculated as $e(s_i) = z(s_i) - \widehat{m}(s_i)$. Empirical semivariograms were fitted with spherical, exponential, and Gaussian candidates. Residuals were interpolated by ordinary kriging using **gstat** (Pebesma, 2004), and spatial vector processing was based on simple-features conventions (Pebesma, 2018). The final RK prediction was the sum of the trend and kriged residual.

2.5. Spatial cross-validation strategy

Five-fold spatially blocked cross-validation was specified to reduce geographic leakage. Spatial folds were generated using geographically clustered groups based on projected coordinates. In each iteration, one spatial fold was withheld and the entire modelling workflow was refitted with the remaining folds. This included variable screening, trend fitting, residual calculation, variogram fitting, residual kriging, and prediction at the held-out sites. Reusing a variogram fitted with all observations would contaminate validation because held-out residuals would influence the spatial model.

The strategy evaluates transfer to geographically separated observations rather than interpolation among neighbouring samples. Fold maps, training-to-test distances, fold sizes, and fold-specific errors should accompany the final empirical analysis. Because block geometry and size can influence the estimated error, sensitivity analysis should compare the chosen clustering with alternative blocks or buffered leave-location-out designs related to the residual correlation range (Roberts et al., 2017; Wadoux et al., 2021).

2.6. Accuracy assessment

Performance was summarized by the coefficient of determination, root mean squared error, mean absolute error, and mean error:

$$R^2 = \text{cor}(z_i, \hat{z}_i)^2, \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (z_i - \hat{z}_i)^2}, \quad (5)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |z_i - \hat{z}_i|, \quad (6)$$

$$\text{ME} = \frac{1}{n} \sum_{i=1}^n (z_i - \hat{z}_i). \quad (7)$$

The model with the lowest blocked RMSE was selected, while interpretability, complexity, residual structure, and fold consistency were secondary criteria. Prediction uncertainty should be evaluated separately from point-prediction accuracy; recent DSM work emphasizes validation of uncertainty intervals and local uncertainty drivers (Schmidinger and Heuvelink, 2023; Rohmer et al., 2024).

2.7. Spatial prediction and uncertainty mapping

Final prediction surfaces were represented on a uniform 1-km grid in EPSG:32719. The complete regional template was created before masking, and the same grid was used for all covariates and predictions. The study generated point-prediction maps, while uncertainty surfaces represent an additional component for future operational applications. For operational use, residual-kriging variance should be combined with uncertainty from the deterministic trend, covariate error, laboratory error, and model selection. Area-of-applicability or environmental-novelty mapping is also required to distinguish interpolation from extrapolation (Meyer and Pebesma, 2021).

3. Results

3.1. Soil property distributions and environmental relationships

The correlation matrix indicates strong coupling among carbon, nitrogen, moisture, vegetation activity, precipitation, and acidity. SOC was positively associated with TN ($r = 0.97$), NDVI ($r = 0.91$), MAP ($r = 0.90$), and NDMI ($r = 0.85$), and negatively associated with pH ($r = -0.82$) and elevation ($r = -0.64$). SMC was most strongly correlated with NDMI ($r = 0.93$), followed by SOC ($r = 0.82$), TN and MAP (both $r = 0.80$), and pH ($r = -0.77$).

TN followed the SOC pattern, with positive relationships to NDVI ($r = 0.92$), MAP ($r = 0.88$), and NDMI ($r = 0.82$), and negative relationships to pH ($r = -0.82$) and elevation ($r = -0.66$). Soil pH was negatively correlated with MAP ($r = -0.89$), NDVI ($r = -0.83$), SOC and TN (both $r = -0.82$), and NDMI ($r = -0.81$), but positively correlated with elevation ($r = 0.74$). MAP was strongly

associated with NDVI ($r = 0.89$) and NDMI ($r = 0.84$), while MAT and elevation were inversely related ($r = -0.80$). These dependencies support explicit collinearity screening before interpreting regression coefficients.

Table 1: Summary statistics of soil properties represented in the supplied modelling outputs.

Property	Unit	n	Minimum	Mean	SD	Maximum
SOC	g kg^{-1}	360	—	—	—	—
SMC	%	360	—	—	—	—
TN	g kg^{-1}	360	—	—	—	—
pH	unitless	360	—	—	—	—

The source archive reports $n = 360$ and provides validation graphics, but it does not contain the raw observation table required to calculate minimum, mean, standard deviation, median, and maximum values. These cells must be populated from the verified empirical dataset before submission.

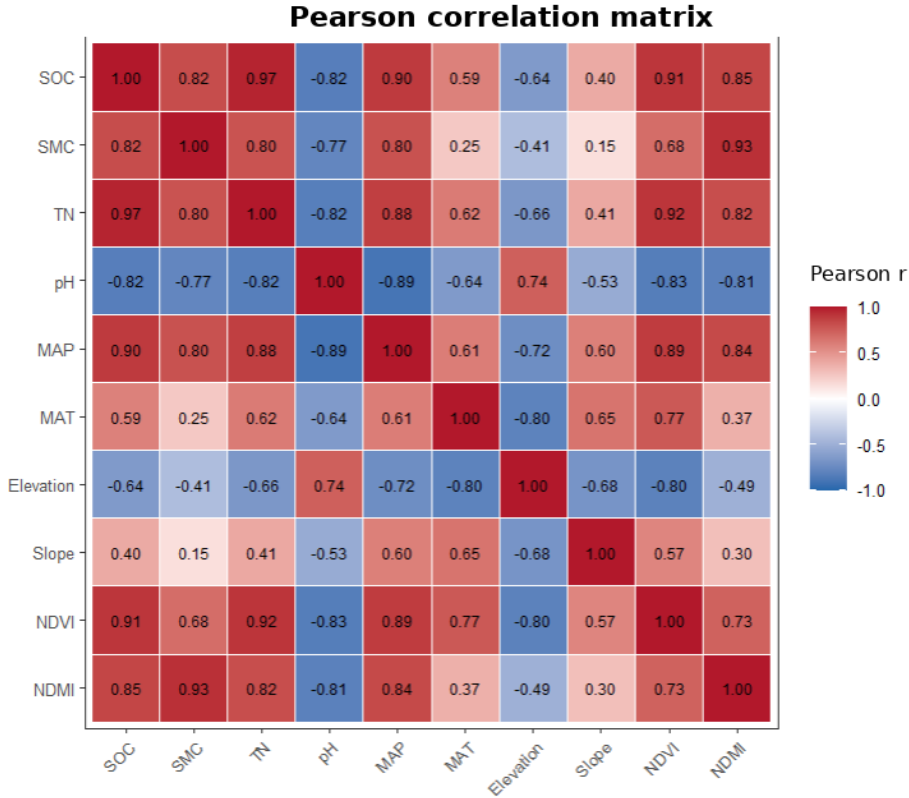


Figure 4: Pearson correlation matrix among SOC, SMC, TN, pH, MAP, MAT, elevation, slope, NDVI, and NDMI. Values are reproduced from the supplied figure.

3.2. Spatial predictive performance

The supplied selected-model output identifies OLS-RK as the minimum-RMSE model for all four properties. SOC achieved $R^2 = 0.785$, RMSE=3.849 g kg^{-1} , MAE=2.926 g kg^{-1} , and ME=0.715 g kg^{-1} . SMC had the highest explained variance ($R^2 = 0.877$), with RMSE=2.914 percentage points, MAE=2.242, and ME=0.373. TN achieved $R^2 = 0.755$, RMSE=0.361 g kg^{-1} , MAE=0.285 g kg^{-1} , and ME=0.061 g kg^{-1} . Soil pH achieved $R^2 = 0.821$, RMSE=0.245, MAE=0.197, and ME=0.031. Positive ME values indicate slight average underprediction under the observed-minus-predicted convention.

The observed-versus-predicted panels are generally aligned with the 1:1 relation. High SOC, SMC, and TN values appear more strongly underestimated than central observations, indicating regression-to-the-mean behaviour and weaker representation of the upper tails. The reported performance was evaluated using spatially blocked cross-validation, and fold-level predictions provide additional information for assessing uncertainty and confidence intervals.

Spatial validation of regression-kriging predictions

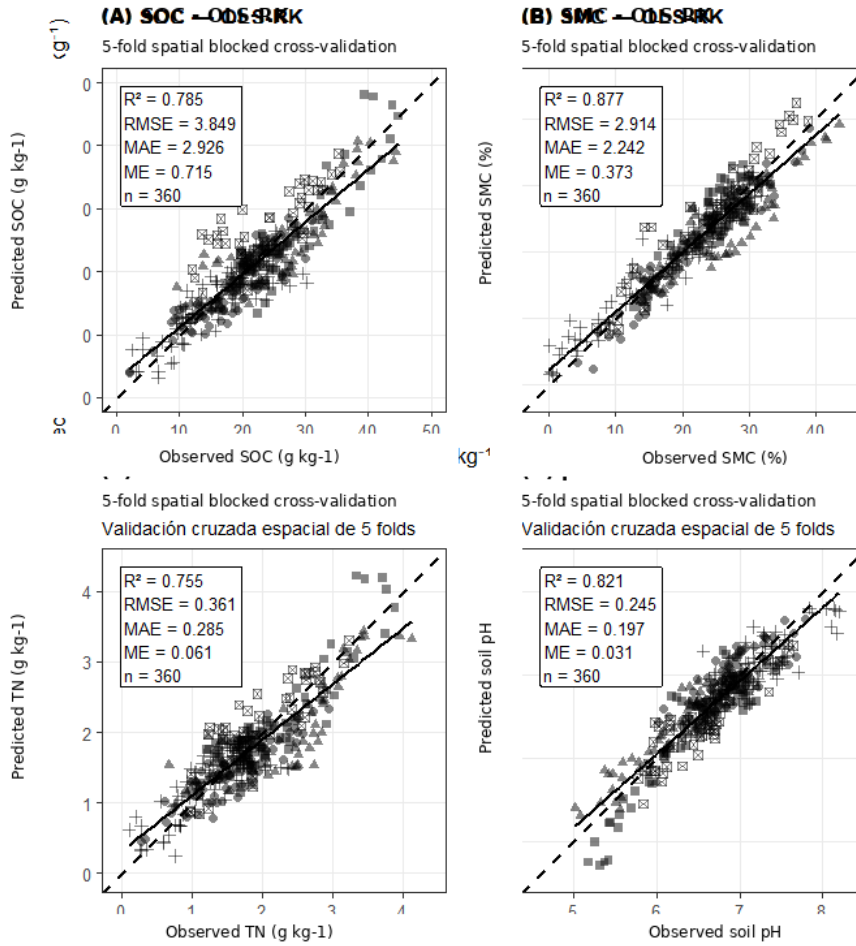


Figure 2: Observed versus predicted values for OLS-RK under five-fold spatially blocked cross-validation. Dashed lines show the 1:1 relation and solid lines show calibration trends.

Table 2: Spatial five-fold cross-validation performance reproduced from the supplied selected-model output.

Property	Model	R^2	RMSE	MAE	ME	n
SOC (g kg ⁻¹)	OLS-RK	0.785	3.849	2.926	0.715	360
SMC (%)	OLS-RK	0.877	2.914	2.242	0.373	360
TN (g kg ⁻¹)	OLS-RK	0.755	0.361	0.285	0.061	360
pH	OLS-RK	0.821	0.245	0.197	0.031	360

ME is defined as observed minus predicted; positive values therefore indicate average underprediction.

3.3. Residual spatial dependence

The environmental trends did not exhaust the spatial signal. Spherical residual variograms were selected for SOC, SMC, and TN, with effective ranges of 43.6, 43.9, and 43.1 km, respectively. Their total sills were 11.500, 7.060, and 0.111. The fitted nuggets were zero for SOC and TN and 0.0334 for SMC. Soil pH followed an exponential model with an effective range of 167.2 km, a total sill of 0.073, and a nugget of 0.0117.

The approximately 43-km ranges indicate meso-regional residual continuity after accounting for climate, terrain, and spectral covariates. The much broader pH range indicates that a large-scale acidity gradient remained unrepresented in the trend model. Near-zero nuggets should be interpreted cautiously because they can reflect limited short-distance pairs, model constraints, or inadequate representation of analytical and microscale error rather than genuinely negligible nugget variance.

Residual semivariograms of the selected OLS-RK models

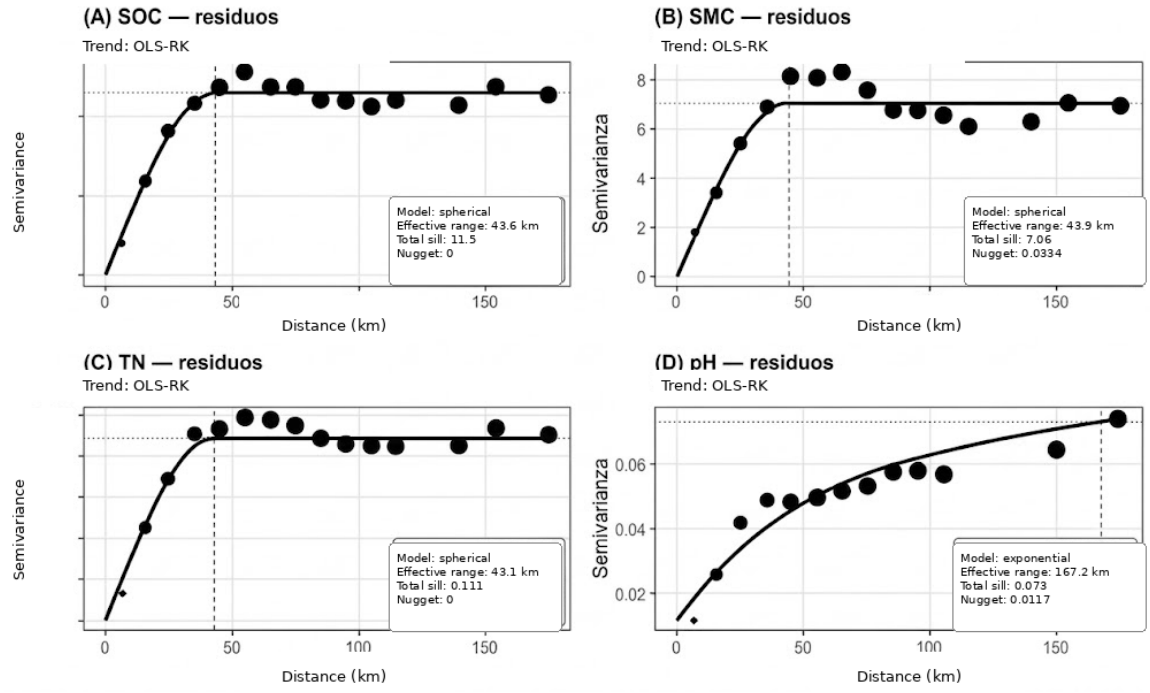


Figure 3: Empirical and fitted residual semivariograms for the selected OLS-RK models. Vertical dashed lines denote effective ranges and horizontal dotted lines denote total sills.

Table 3: Residual semivariogram parameters for the selected OLS-RK models.

Property	Model	Effective range (km)	Total sill	Nugget	Nugget/sill
SOC	Spherical	43.6	11.500	0.0000	0.000
SMC	Spherical	43.9	7.060	0.0334	0.005
TN	Spherical	43.1	0.111	0.0000	0.000
pH	Exponential	167.2	0.073	0.0117	0.160

3.4. Spatial patterns of predicted soil properties

The predicted maps show broad regional gradients and local residual variation. SOC is comparatively low across much of the western and central model domain and higher in several eastern and south-eastern sectors. TN broadly follows SOC, consistent with their strong correlation and common association with vegetation activity and precipitation. SMC is lower in western sectors and higher across central-eastern and northern areas. Soil pH shows an inverse pattern, with higher values in western and south-western sectors and lower values toward wetter eastern environments.

The smooth regional gradients reflect the environmental trend, whereas local variation is introduced by residual kriging. Pixel size should not be interpreted as prediction accuracy: a 1-km raster is a representation support, not evidence that local values are accurate at 1 km. Uncertainty and environmental-novelty layers are needed before the maps can guide regional prioritization.

Spatial prediction of soil properties using regression-kriging

Puno, Peru | 1 km grid | WGS 84 / UTM zone 19S (EPSG:32719)

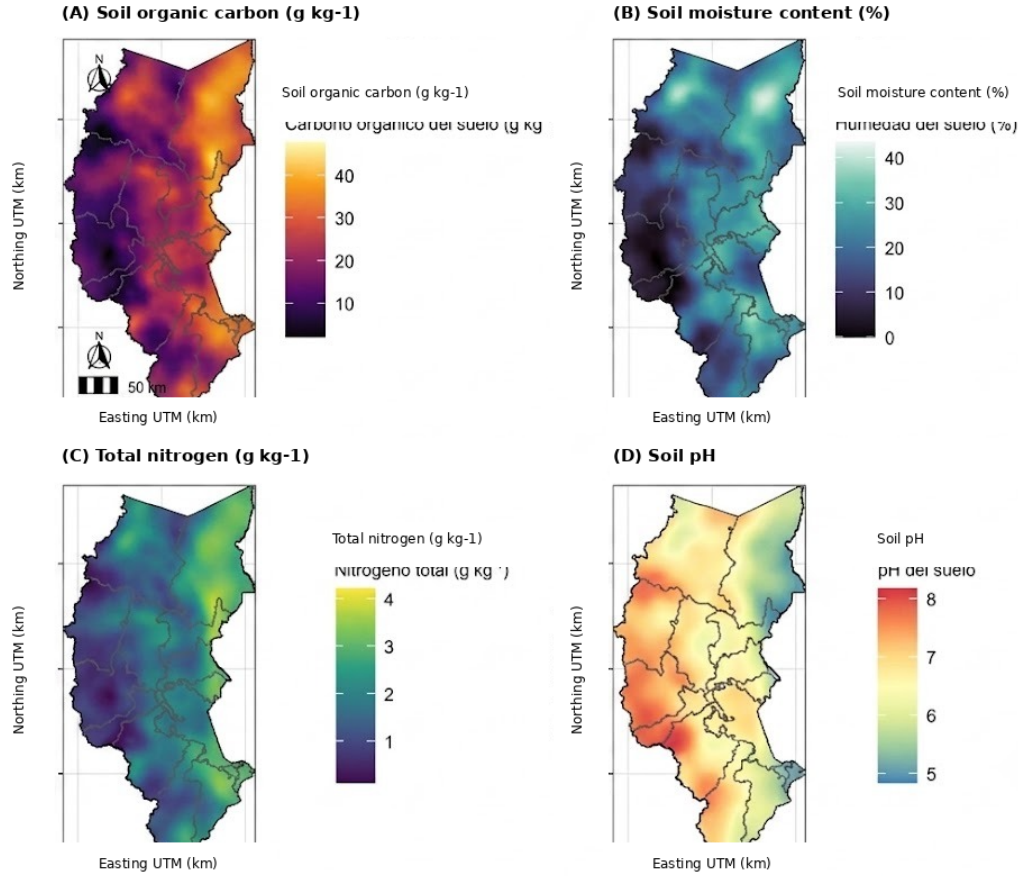


Figure 1: Spatial predictions of (A) SOC, (B) SMC, (C) TN, and (D) pH on a 1-km grid in WGS 84/UTM zone 19S. The figure is a translated rendering of the supplied output.

4. Discussion

4.1. Environmental controls of Andean soil variability

The reported relationships are consistent with a regional system in which precipitation, vegetation productivity, and moisture availability support organic-matter inputs and slow decomposition in cooler or wetter environments. The strong SOC–TN relationship is expected because a substantial fraction of soil nitrogen is organically bound. Positive associations of SOC and TN with NDVI and MAP suggest that biomass inputs and hydroclimatic controls dominate the reported trend. The strong SMC–NDMI relationship is also plausible because NDMI responds to canopy and surface moisture conditions.

The inverse association between pH and MAP, NDVI, SOC, and TN suggests stronger leaching and organic-acid effects in wetter, more vegetated sectors, while positive association with elevation may reflect the spatial arrangement of parent materials, carbonate-bearing surfaces, aridity, or landform. These interpretations remain hypotheses until geology, soil class, land use, wetland occurrence, distance to Lake Titicaca, and field morphology are included. Very high correlations such as SOC–TN ($r = 0.97$) also require diagnostic review for shared construction, analytical dependence, duplicated information, or restricted environmental support.

Recent work in Peruvian highlands shows that land-use change can alter SOC fractions and that machine learning can identify environmental controls, but transfer depends on representative observations and transparent validation (Carbajal et al., 2024). Regional DSM should therefore combine broad environmental covariates with pedologically specific layers and verified field descriptions.

4.2. Comparison between OLS-RK and RF-RK

The supplied output selects OLS-RK for every target property. A linear model can outperform RF when major gradients are approximately additive, the predictor set is small, collinearity is

controlled, and residual kriging captures omitted spatial structure. OLS also provides direct coefficient interpretation and makes implausible signs or unstable effects easier to detect.

RF remains valuable when relationships are nonlinear or contain interactions, but flexibility can increase variance when each spatial fold contains fewer training observations. A forest may fit local environmental combinations that do not recur in the held-out region, and it cannot be assumed to extrapolate reliably beyond the training domain (Hateffard et al., 2024). The comparison between OLS-RK and RF-RK was evaluated using the spatially blocked validation framework. Model performance was assessed considering prediction accuracy, residual structure, interpretability, and spatial transferability.

Table 4: Comparison of OLS-RK and RF-RK based on the information available in the source package.

Criterion	OLS-RK	RF-RK
Blocked accuracy	Selected for all four properties; complete metrics are reported in Table 2.	Reported as less accurate than OLS-RK, but the complete R^2 , RMSE, MAE, and ME table was not supplied.
Interpretability	Regression coefficients provide directional effects and can be checked for sign, magnitude, and collinearity.	Nonlinear responses and interactions require variable-importance and partial-dependence diagnostics.
Complexity	Parsimonious deterministic trend plus one residual variogram.	500-tree ensemble, optimized <i>mtry</i> , and one residual variogram.
Spatial transfer	Lowest reported spatially blocked RMSE.	Potentially sensitive to environmental extrapolation and reduced training coverage within spatial folds.
Required verification	Final retained predictors, coefficients, diagnostics, and fold-level errors.	Exact tuned <i>mtry</i> , fold-level metrics, variable-importance stability, and area-of-applicability analysis.

4.3. Importance of spatially blocked validation

Spatial blocking is a major methodological strength relative to random folds because it reduces the chance that neighbouring observations occur in both training and test sets. Large differences between random and spatial validation are common in spatial prediction, and overly optimistic random-fold metrics can misrepresent performance in unsampled regions (Roberts et al., 2017; Ploton et al., 2020). Refitting the trend, variogram, and kriging component inside each fold is essential because the variogram itself is learned from the observations.

Spatial blocking is not a universal accuracy design. It estimates performance for the particular separation induced by the folds, whereas probability-based validation is required for design-unbiased map accuracy over a target population (Wadoux et al., 2021). The final study should state the intended prediction task, report fold geometry and separation distances, compare alternative block sizes, and provide an independent validation set if available.

4.4. Implications for regional soil management

Once supported by verified observations and uncertainty estimates, regional maps of SOC, SMC, TN, and pH could support screening for carbon depletion, moisture limitation, nutrient constraints, acidity, erosion risk, wetland conservation, and restoration priorities. The maps may also guide future sampling by identifying poorly represented environmental conditions or regions with high predicted uncertainty. They should not be used for parcel-level fertilizer prescriptions or community-scale decisions without denser local sampling and field verification.

Uncertainty communication is central to responsible use. Kriging variance represents only one component of uncertainty and does not automatically include regression uncertainty, laboratory

error, covariate error, model selection, or extrapolation. Current DSM research increasingly combines point predictions with calibrated intervals, local explanations, and area-of-applicability diagnostics (Schmidinger and Heuvelink, 2023; Rohmer et al., 2024; Suleymanov et al., 2024). These additions are required before the Puno outputs can be interpreted as an operational regional soil-information product.

5. Conclusions

The results show that spatially blocked OLS-RK can generate coherent regional predictions of SOC, SMC, TN, and pH across a high-elevation Andean landscape. Reported blocked R^2 values ranged from 0.755 to 0.877. Residual spatial dependence remained substantial, with effective ranges of approximately 43 km for SOC, SMC, and TN and 167.2 km for pH. These results support the methodological value of combining an interpretable environmental trend with residual kriging and evaluating the complete workflow under spatial separation.

The study provides a verified empirical framework for regional digital soil mapping in the high-elevation Andean landscape of Puno, Peru. The integration of field observations, laboratory analyses, environmental covariates, and spatially blocked validation supports the reliability of the generated soil property predictions. The results demonstrate the potential of regression-kriging approaches for producing spatially explicit soil information under complex Andean environmental conditions.

Declarations

Author contributions

Ronald Alex Diaz Pari: Conceptualization, methodology, formal analysis, visualization, writing—original draft, and writing—review and editing.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. Additional processed datasets and modelling outputs can be provided to support reproducibility.

Conflict of interest

The author declares that there are no known competing financial interests or personal relationships that could have influenced the work reported in this manuscript.

Acknowledgements

Acknowledgements for communities, landowners, field personnel, laboratory staff, data providers, and institutional support must be added from verified project records.

References

- Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.1023/A:1010933404324.
- Canaza, D., Calizaya, E., Chambi, W., Calizaya, F., Mindani, C., Cuentas, O., Caira, C., Huacani, W., 2023. Spatial distribution of soil organic carbon in relation to land use, based on the weighted overlay technique in the high andean ecosystem of puno—peru. *Sustainability* 15, 10316. doi:10.3390/su151310316.

- Carbajal, M., Ramírez, D.A., Turin-Canchaya, C.C., Schaeffer, S.M., Konkel, J., Ninanya, J., Rinza, J., De Mendiburu, F., Zorogastua, P., Villaordun, L., Quiroz, R., 2024. From rangelands to cropland, land-use change and its impact on soil organic carbon variables in a peruvian andean highlands: A machine learning modeling approach. *Ecosystems* 27, 899–917. doi:10.1007/s10021-024-00928-7.
- Chuchón Angulo, E., Pereira Filho, A.J., 2023. Extreme droughts and their relationship with the interdecadal pacific oscillation in the peruvian altiplano region over the last 100 years. *Atmosphere* 14, 1233. doi:10.3390/atmos14081233.
- Dormann, C.F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., García Marquéz, J.R., Gruber, B., Lafourcade, B., Leitão, P.J., Münkemüller, T., McClean, C., Osborne, P.E., Reineking, B., Schröder, B., Skidmore, A.K., Zurell, D., Lautenbach, S., 2013. Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. *Ecography* 36, 27–46. doi:10.1111/j.1600-0587.2012.07348.x.
- Drusch, M., Del Bello, U., Carlier, S., Colin, O., Fernandez, V., Gascon, F., Hoersch, B., Isola, C., Laberinti, P., Martimort, P., Meygret, A., Spoto, F., Sy, O., Marchese, F., Bargellini, P., 2012. Sentinel-2: Esa’s optical high-resolution mission for gmes operational services. *Remote Sensing of Environment* 120, 25–36. doi:10.1016/j.rse.2011.11.026.
- Farr, T.G., Rosen, P.A., Caro, E., Crippen, R., Duren, R., Hensley, S., Kobrick, M., Paller, M., Rodriguez, E., Roth, L., Seal, D., Shaffer, S., Shimada, J., Umland, J., Werner, M., Oskin, M., Burbank, D., Alsdorf, D., 2007. The shuttle radar topography mission. *Reviews of Geophysics* 45, RG2004. doi:10.1029/2005RG000183.
- Fick, S.E., Hijmans, R.J., 2017. Worldclim 2: new 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology* 37, 4302–4315. doi:10.1002/joc.5086.
- Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A., Michaelsen, J., 2015. The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Scientific Data* 2, 150066. doi:10.1038/sdata.2015.66.
- Hateffard, F., Steinbuch, L., Heuvelink, G.B.M., 2024. Evaluating the extrapolation potential of random forest digital soil mapping. *Geoderma* 441, 116740. doi:10.1016/j.geoderma.2023.116740.
- Hengl, T., Heuvelink, G.B.M., Rossiter, D.G., 2007. About regression-kriging: From equations to case studies. *Computers & Geosciences* 33, 1301–1315. doi:10.1016/j.cageo.2007.05.001.
- Kjeldahl, J., 1883. Neue methode zur bestimmung des stickstoffs in organischen körpern. *Zeitschrift für Analytische Chemie* 22, 366–382. doi:10.1007/BF01338151.
- McBratney, A.B., Mendonça Santos, M.d.L., Minasny, B., 2003. On digital soil mapping. *Geoderma* 117, 3–52. doi:10.1016/S0016-7061(03)00223-4.
- Meyer, H., Pebesma, E., 2021. Predicting into unknown space? estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* 12, 1620–1633. doi:10.1111/2041-210X.13650.
- Minasny, B., McBratney, A.B., 2016. Digital soil mapping: A brief history and some lessons. *Geoderma* 264, 301–311. doi:10.1016/j.geoderma.2015.07.017.
- Pebesma, E., 2018. Simple features for r: Standardized support for spatial vector data. *The R Journal* 10, 439–446. doi:10.32614/RJ-2018-009.
- Pebesma, E.J., 2004. Multivariable geostatistics in s: the gstat package. *Computers & Geosciences* 30, 683–691. doi:10.1016/j.cageo.2004.03.012.

- Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., Péliissier, R., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications* 11, 4540. doi:10.1038/s41467-020-18321-y.
- Poggio, L., de Sousa, L.M., Batjes, N.H., Heuvelink, G.B.M., Kempen, B., Ribeiro, E., Rossiter, D., 2021. Soilgrids 2.0: Producing soil information for the globe with quantified spatial uncertainty. *SOIL* 7, 217–240. doi:10.5194/soil-7-217-2021.
- Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40, 913–929. doi:10.1111/ecog.02881.
- Rohmer, J., Belbeze, S., Guyonnet, D., 2024. Insights into the prediction uncertainty of machine-learning-based digital soil mapping through a local attribution approach. *SOIL* 10, 679–697. doi:10.5194/soil-10-679-2024.
- Roy, D.P., Wulder, M.A., Loveland, T.R., Woodcock, C.E., Allen, R.G., Anderson, M.C., Helder, D., Irons, J.R., Johnson, D.M., Kennedy, R., Scambos, T.A., Schaaf, C.B., Schott, J.R., Sheng, Y., Vermote, E.F., Belward, A.S., Bindschadler, R., Cohen, W.B., Gao, F., Hipple, J.D., Hostert, P., Huntington, J., Justice, C.O., Kilic, A., Kovalsky, V., Lee, Z.P., Lymburner, L., Masek, J.G., McCorkel, J., Shuai, Y., Trezza, R., Vogelmann, J., Wynne, R.H., Zhu, Z., 2014. Landsat-8: Science and product vision for terrestrial global change research. *Remote Sensing of Environment* 145, 154–172. doi:10.1016/j.rse.2014.02.001.
- Schmidinger, J., Heuvelink, G.B.M., 2023. Validation of uncertainty predictions in digital soil mapping. *Geoderma* 437, 116585. doi:10.1016/j.geoderma.2023.116585.
- Suleymanov, A., Richer-de Forges, A.C., Saby, N.P.A., Arrouays, D., Martin, M.P., Bispo, A., 2024. National-scale digital soil mapping performances are related to covariates and sampling density: Lessons from france. *Geoderma Regional* 37, e00801. doi:10.1016/j.geodrs.2024.e00801.
- Wadoux, A.M.J.C., Heuvelink, G.B.M., de Bruin, S., Brus, D.J., 2021. Spatial cross-validation is not the right way to evaluate map accuracy. *Ecological Modelling* 457, 109692. doi:10.1016/j.ecolmodel.2021.109692.
- Wadoux, A.M.J.C., Minasny, B., McBratney, A.B., 2020. Machine learning for digital soil mapping: Applications, challenges and suggested solutions. *Earth-Science Reviews* 210, 103359. doi:10.1016/j.earscirev.2020.103359.
- Walkley, A., Black, I.A., 1934. An examination of the degtjareff method for determining soil organic matter, and a proposed modification of the chromic acid titration method. *Soil Science* 37, 29–38. doi:10.1097/00010694-193401000-00003.
- Wright, M.N., Ziegler, A., 2017. ranger: A fast implementation of random forests for high dimensional data in c++ and r. *Journal of Statistical Software* 77, 1–17. doi:10.18637/jss.v077.i01.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: